

Econometrics

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Week 6 Notes

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1 Simultaneous Equations Models

1.1 Introduction

In regression models one of the crucial assumptions we make is that the exogenous variables $x's$ are independent of the error term, however this assumption is violated under simultaneous equations.

Simultaneous equations are models with more than one response variable where the solution is determined by an equilibrium among opposing forces.

1.1.1 Examples of SEM

A very simple supply and demand model might look like

$$\text{Demand} : Q = \alpha_1 P + \alpha_2 X + e_d$$

$$\text{Supply} : Q = \beta_1 P + e_s$$

In the above model P and Q are called endogenous variables because their values are determined within the system we have created.

The income variable X has a value that is determined outside this system. Such variables are said to be exogenous and these variables are treated like usual X explanatory variables.

The model of money wage and price determination is a good example of SEM. According to the model the wage rate is determined by unemployment rate in the economy and the rate of price change in an unrestricted labour market. The rate of inflation on the other hand is a function of the change in the wage rate and money supply. The model contains the equations:

$$W_t = \alpha_0 + \alpha_1 U_t + \alpha_2 P_t + u_{1t}$$

$$P_t = \beta_0 + \beta_1 W_t + \beta_2 R_t + \beta_3 M_t + u_{2t}$$

W is rate of change in money wage, U unemployment rate in percentage, P rate of change in prices, R rate of change in cost capital, M supply of money, t is time and u_1, u_2 stochastic disturbances.

1.2 Endogenous and Exogenous Variables

-Endogenous variables get determined jointly and interdependently within the model. The endogenous variables are determined by the exogenous variables.

- Exogenous variables are determined outside the model and are independently of the endogenous variable. These exogenous variables determine the endogenous variable but the endogenous variables cannot affect exogenous variables
- Lagged dependent variables: in a model we can include the lagged values of an endogenous variable also.

The difference between single equation model and SEM is given in the table below:

Single Equation Model	Simultaneous Equation Model
All independent variables are exogenous	Independent variables can be both exogenous and lagged endogenous
Equation of the model can be solved independently	To solve an equation of the model we have to consider other equations in the system
Dataset represents a specific equation	The exact form of the equation cannot be specified from data set
No need to identify an equation as the functional form of the equation is specified	Identification is required before solving the equations
OLS estimation is consistent	OLS estimators are inconsistent
Solution technique: OLS, GLS	Solution techniques: ILS, 2SLS Instrumental variable method 3SLS

Applying OLS to a system of simultaneous equations we get estimates that are not unbiased and not consistent.

In order to obtain unbiased and consistent estimates we apply any of the following methods

- Reduced form method,
- The method of instrumental variables
- Two-stage least squares
- Limited Information maximum likelihood
- Three stage least squares
- Full information maximum likelihood

1.3 Structural and Reduced Form Equations

An economic model such as demand and supply equations include several of the dependent variables in each equation such a model is called **structural form of the model**

If the structural form is transformed such that each equation shows one dependent variable as a function of only exogenous independent variable the new form is called **reduced form**

The reduced form can be estimated by least squares while the structural form cannot because it includes exogenous variables on its RHS.

The necessary condition for the identification requires that for the problem to have a solution each equation in structural form of the system should miss at least an exogenous variable that is present in other equations.

We consider the demand equations:

$$Q = \alpha_1 P + \alpha_2 X + e_d$$

$$Q = \beta_1 P + e_s$$

To solve for P set Q in the demand and supply equations to be equal

$$\beta_1 P + e_s = \alpha_1 P + \alpha_2 X + e_d$$

Solving for P

$$P = \frac{\alpha_2}{\beta_1 - \alpha_1} X + \frac{e_d - e_s}{\beta_1 - \alpha_1}$$

Which can be expressed as:

$$P = \pi_1 X + v_1$$

Solving for Q:

$$Q = \beta_1 P + e_s = \frac{\beta_1 \alpha_2}{\beta_1 - \alpha_1} X + \frac{\beta_1 e_d - e_s}{\beta_1 - \alpha_1} = \pi_2 X + v_2$$

The parameters π_1 and π_2 are called reduced form equations and the error terms v_1 and v_2 are called reduced form errors.

The least squares estimators of the parameters in a SEM is biased and inconsistent because of the correlation between the random error and endogenous variables on the RHS of the equation.

The necessary condition for identification requires that for the problem to have a solution each equation in structural form of the system should miss at least an exogenous variable that is present in other equation.

1.4 Two stage least squares estimation

The most widely used method for estimating the parameters of an identified structural equation is called two-stage least squares.

We consider the supply equation discussed previously. We cannot apply the usual least squares procedures to estimate β_1 in this equation because the endogenous variable P on the RHS of the equation is correlated with error term e_s .

The reduced form model is:

$$P = \pi_1 X + v_1$$

Suppose we know π_1 then

$$Q = \beta_1 E[p] + (\beta_1 v_1 + e_s)$$

We can estimate π_1 using $\hat{\pi}_1$ from the reduced form equation for P. A consistent estimator for $E[P]$ is

$$\hat{P} = \hat{\pi}_1 X$$

Estimating the above equation generates the so called two stage least squares estimate for β_1 which is consistent and normally distributed in large sample.

The 2SLS estimation procedures are:

- Least square estimation of the reduced form equation for P and the calculation of its predicted value $\hat{P}$
- Least square estimation of SE in which RHS endogenous variable is replaced by its predicted value $\hat{P}$

The properties of the two stage least squares are as follows:

- The 2SLS estimator is a biased but it is consistent
- In large samples the 2SLS estimator is approximately normally distributed.

2 Examples in R

2.1 Example 1

Simultaneous equation are object of package `systemfit` in R with function `systemfit()` which requires:

- `formula` list describing equations of the system
- `method` desired method of estimation
- `inst` list of instrumental variables

Consider the supply and demand models for truffles:

Demand equation:

$$Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 P S_i + \alpha_4 D I_i + e_i^d$$

The supply equation:

$$Q_i = \beta_1 + \beta_2 P_i + \beta_3 P F_i + e_i^s$$

Quantity and price are endogenous and all other variables are considered exogenous.

The reduced form equations are:

$$Q_i = \pi_{11} + \pi_{21} P S_i + \pi_{31} D I_i + \pi_{41} P F_i + V_{i1}$$

$$P_i = \pi_{12} + \pi_{22} P S_i + \pi_{32} D I_i + \pi_{42} P F_i + v_{i2}$$

```

library(systemfit)
data("truffles", package="PoEdata")
D <- q~p+ps+di
S <- q~p+pf
sys <- list(D,S)
instr <- ~ps+di+pf
truff.sys <- systemfit(sys, inst=instr,
                      method="2SLS", data=truffles)
summary(truff.sys)

##
## systemfit results
## method: 2SLS
##
##          N DF      SSR detRCov   OLS-R2 McElroy-R2
## system 60 53 692.472 49.8028 0.438964   0.807408
##
##          N DF      SSR      MSE   RMSE      R2   Adj R2
## eq1 30 26 631.9171 24.30450 4.92996 -0.023950 -0.142098
## eq2 30 27  60.5546  2.24276 1.49758  0.901878  0.894610
##
## The covariance matrix of the residuals
##          eq1      eq2
## eq1 24.30451 2.16943
## eq2  2.16943 2.24276
##
## The correlations of the residuals
##          eq1      eq2
## eq1 1.00000 0.29384
## eq2 0.29384 1.00000
##
##
## 2SLS estimates for 'eq1' (equation 1)
## Model Formula: q ~ p + ps + di
## Instruments: ~ps + di + pf
##
##          Estimate Std. Error  t value  Pr(>|t|)
## (Intercept) -4.279471   5.543884 -0.77193 0.4471180
## p           -0.374459   0.164752 -2.27287 0.0315350 *
## ps           1.296033   0.355193  3.64881 0.0011601 **
## di           5.013977   2.283556  2.19569 0.0372352 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.92996 on 26 degrees of freedom
## Number of observations: 30 Degrees of Freedom: 26
## SSR: 631.917143 MSE: 24.304505 Root MSE: 4.92996
## Multiple R-Squared: -0.02395 Adjusted R-Squared: -0.142098
##
##
## 2SLS estimates for 'eq2' (equation 2)
## Model Formula: q ~ p + pf
## Instruments: ~ps + di + pf
##

```

```
##               Estimate Std. Error  t value  Pr(>|t|)
## (Intercept) 20.0328022  1.2231148  16.3785 1.5543e-15 ***
## p           0.3379816  0.0249196  13.5629 1.4344e-13 ***
## pf          -1.0009094  0.0825279 -12.1281 1.9456e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.497585 on 27 degrees of freedom
## Number of observations: 30 Degrees of Freedom: 27
## SSR: 60.554565 MSE: 2.242762 Root MSE: 1.497585
## Multiple R-Squared: 0.901878 Adjusted R-Squared: 0.89461
```

The output of the `systemfit()` function shows the estimates by structural equation: `eq1` is the demand function, where, as expected, price has a negative sign, and `eq2` is the supply equation, with a positive sign for price.

By evaluating the reduced form equation using OLS, one can determine the effects of changes in exogenous variables on the equilibrium market price and quantity, while the structural equations show the effects of such changes on the quantity demanded, respectively on the quantity supplied.

Estimating the structural equations by such methods as 2SLS is, in fact, estimating the market demand and supply curves, which is extremely useful for economic analysis.

Estimating the reduced forms, while being useful for prediction, does not allow for deep analysis - it only gives the equilibrium point, not the whole curves.

```
# reduced form equation for demand
Q.red <- lm(q~ps+di+pf, data=truffles)

# reduced form equation for price
P.red <- lm(q~ps+di+pf, data=truffles)
```

The results show that all the exogenous variables have significant effects on the equilibrium quantity and price and have the expected signs.

2.2 Example 2

The `fultonfish` dataset provides another demand and supply example where the simultaneous equations method can be applied.

The purpose of this example is to emphasize that the exogenous variables that are key for identification must be statistically significant

The equations are:

$$\log(\text{quan}) = \alpha_1 + \alpha_2 \log(\text{price}) + \alpha_3 \text{mon} + \alpha_4 \text{tue} + \alpha_5 \text{wed} + \alpha_6 \text{thus} + e_D$$

$$\log(\text{quan}) = \beta_1 + \beta_2 \log(\text{price}) + \beta_3 \text{stormy} + e_S$$

the endogenous variables are `lprice`, the log of price, and `lquan` the exogenous variables are the indicator variables for the day of the week, and whether the catching day was stormy.

The reduced form equations are:

$$\log(q) = \pi_{11} + \pi_{21} \text{mon} + \pi_{31} \text{tue} + \pi_{41} \text{wed} + \pi_{51} \text{thu} + \pi_{61} \text{stormy} + v_1$$

$$\log(p) = \pi_{12} + \pi_{22}mon + \pi_{32}tue + \pi_{42}wed + \pi_{52}thu + \pi_{62}stormy + v_2$$

```
data("fultonfish", package="PoEdata")
fish.D <- lquan~lprice+mon+tue+wed+thu
fish.S <- lquan~lprice+stormy
fish.eqs <- list(fish.D, fish.S)
fish.ivs <- ~mon+tue+wed+thu+stormy
fish.sys <- systemfit(fish.eqs, method="2SLS",
                     inst=fish.ivs, data=fultonfish)
summary(fish.sys)

##
## systemfit results
## method: 2SLS
##
##          N DF      SSR detRCov   OLS-R2 McElroy-R2
## system 222 213 109.612 0.107301 0.094242 -0.597812
##
##          N DF      SSR      MSE      RMSE      R2  Adj R2
## eq1 111 105 52.0903 0.496098 0.704342 0.139124 0.098130
## eq2 111 108 57.5218 0.532610 0.729801 0.049360 0.031755
##
## The covariance matrix of the residuals
##          eq1      eq2
## eq1 0.496098 0.396138
## eq2 0.396138 0.532610
##
## The correlations of the residuals
##          eq1      eq2
## eq1 1.000000 0.770653
## eq2 0.770653 1.000000
##
##
## 2SLS estimates for 'eq1' (equation 1)
## Model Formula: lquan ~ lprice + mon + tue + wed + thu
## Instruments: ~mon + tue + wed + thu + stormy
##
##          Estimate Std. Error  t value  Pr(>|t|)
## (Intercept)  8.5059113  0.1661669  51.18896 < 2.22e-16 ***
## lprice       -1.1194169  0.4286450  -2.61152  0.0103334 *
## mon          -0.0254022  0.2147742  -0.11827  0.9060766
## tue          -0.5307694  0.2080001  -2.55177  0.0121574 *
## wed          -0.5663511  0.2127549  -2.66199  0.0089895 **
## thu           0.1092673  0.2087866   0.52334  0.6018373
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.704342 on 105 degrees of freedom
## Number of observations: 111 Degrees of Freedom: 105
## SSR: 52.090321 MSE: 0.496098 Root MSE: 0.704342
## Multiple R-Squared: 0.139124 Adjusted R-Squared: 0.09813
##
##
```

```
## 2SLS estimates for 'eq2' (equation 2)
## Model Formula: lquan ~ lprice + stormy
## Instruments: ~mon + tue + wed + thu + stormy
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.62835440  0.38897023 22.18256 < 2e-16 ***
## lprice       0.00105931  1.30954697  0.00081  0.99936
## stormy      -0.36324606  0.46491248 -0.78132  0.43632
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.729801 on 108 degrees of freedom
## Number of observations: 111 Degrees of Freedom: 108
## SSR: 57.521843 MSE: 0.53261 Root MSE: 0.729801
## Multiple R-Squared: 0.04936 Adjusted R-Squared: 0.031755
```

The reduced equations are:

```
# q
fishQ.ols <- lm(lquan~mon+tue+wed+thu+stormy, data=fultonfish)

# p
fishP.ols <- lm(lprice~mon+tue+wed+thu+stormy, data=fultonfish)
```